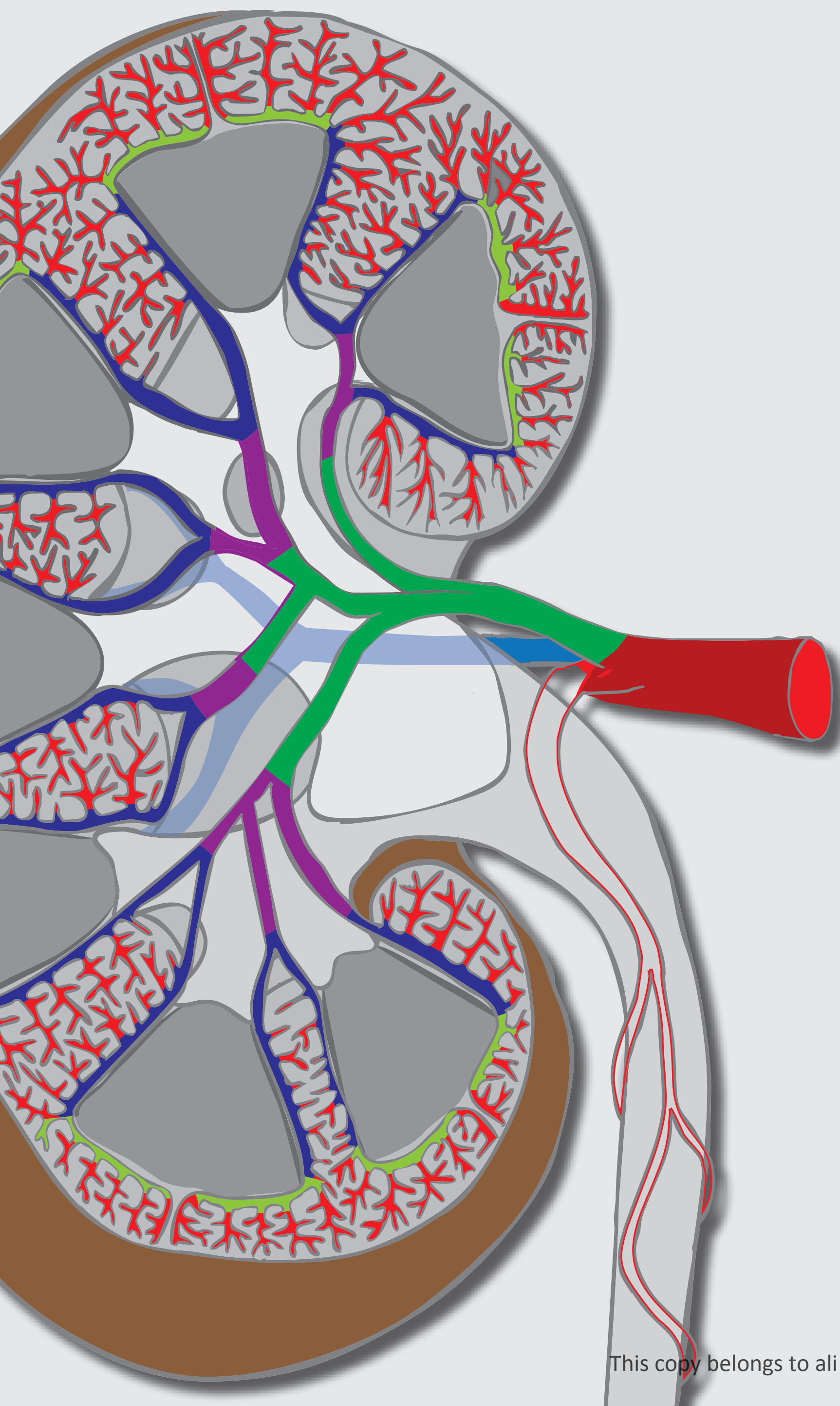
7 Urinary system

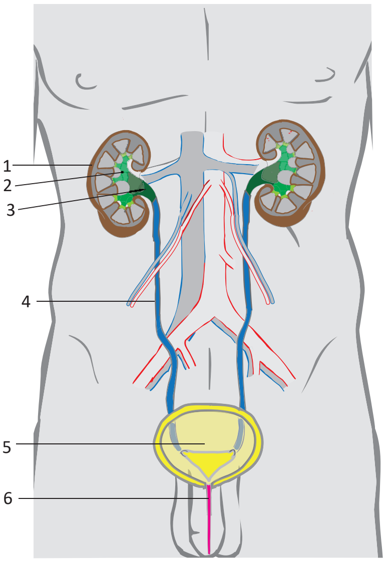
Barbora Beňová
David Kachlík
Radovan Hudák
Ondřej Volný
Adam Whitley



The urinary system is comprised of **two kidneys, two renal pelves, two ureters, the urinary bladder and the urethra**. The kidney is the main organ of the urinary system. Its roles include **plasma filtration, excretion of the waste products of metabolism, acid-base homeostasis**, **regulation of blood pressure and blood volume** (via renin), **hormone secretion** (erythropoetin) and **metabolism of vitamin D**. Urine is transported via the renal pelves and ureters to the urinary bladder before being excreted through the urethra.

External structure

- 1 **Kidneys** (*renes*)
  – filter plasma to create urine
- 2 **Minor and major renal calices** (*calices renales majores et minores*)
  – direct urine from the renal pyramids to the renal pelvis
- 3 **Renal pelvis** (*pelvis renalis*)
  – collects urine and is continuous with the ureter
- 4 **Ureter**
  – the longest segment of the urinary system
  – located between the renal pelvis and urinary bladder
- 5 **Urinary bladder** (*vesica urinaria*)
  – a hollow organ lying behind the pubic symphysis
- 6 **Urethra**
  – the terminal segment of the urinary system
  – **female urethra** (*urethra feminina*)
  – **male urethra** (*urethra masculina*)



Internal structure

Mucosa (*tunica mucosa*) – is lined with transitional epithelium (urothelium), except for at the terminal part of the urethra, which is lined with stratified squamous epithelium

Muscular coat (*tunica muscularis*)
– is organised in two layers: an external circular layer and an internal longitudinal layer
– the muscular coat of the urinary bladder has three layers

Serosa (*tunica serosa*) – peritoneum covers the cranial surface of the urinary bladder and forms a pouch between the posterior wall of the urinary bladder and uterus in the female and between the urinary bladder and rectum in the male

Adventitia (*tunica adventitia*) – loose connective tissue surrounding parts of the urinary system not covered by the peritoneum

Development

Kidney
– originates from the intermediate mesoderm in the pelvic region
– during prenatal development, the caudal half of the body grows faster than the cranial half, which causes the kidneys to appear to ascend relative to their surroundings
– proceeds through three developmental stages

1. **Pronephros** – the first developmental stage
   – disappears completely by the 4th week of prenatal life
   – the mesonephric duct of Wolff / Wolffian duct (*ductus mesonephricus Wolffi*) arises from the caudal part of the pronephric duct
2. **Mesonephros** – arises from the mesonephric duct
   – in males, begins to disappear during the 9th week of prenatal life and consequently transforms into the rete testis and efferent ductules *(ductuli efferentes testis)*
   – in females, disappears along with the mesonephric duct
3. **Metanephros** – the definite kidney, distinguished from the previous stages by the presence of the loop of Henle and the renal medulla, which are capable creating concentrated urine

Renal calices, renal pelvis and ureters
– **the ureteric diverticulum** (*diverticulum ureteris*) arises from the mesonephric duct and develops into the ureter, renal pelvis, calices, papillary and collecting ducts

Urinary bladder and urethra
– the anterior part of **the urogenital sinus** (*sinus urogenitalis*) transforms into the urinary bladder and several parts of the urethra (intramural, prostatic and intermediate parts)

The excretory system is a synonymous term the urinary system. However, as other organ systems take part in the excretion of waste products, this term is less accurate.

Nephros *(pl. nephroi)* is the Greek term for kidney.

The urogenital sinus is the anterior half of the cloaca. The cloaca is the common end of the urinary, digestive and genital systems in the developing human. It divides into the urogenital sinus and anal canal.

Primary urine is plasma ultrafiltrate. 150–180 l of primary urine are produced per day.

There are approximately 1 to 1.5 million nephrons in a single kidney.

Clinical notes

Developmental variations in the shape and size of the kidney (that may or may not result in malfunction):
– **horseshoe kidney** *(ren arcuatus/unguliformis)*
– **supernumerary kidney** *(ren supernumerarius)*
– **sigmoid kidney** *(ren sigmoideus)*
– **cake (lump/fused pelvic) kidney** *(ren fungiformis)*

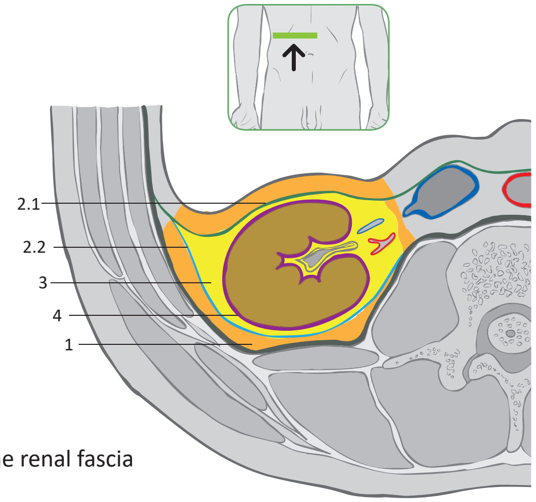
Dissection of the renal fibrous capsule is a procedure that is infrequently performed after the resection of benign tumors to cover the resulting defect.

Tapottement is part of the physical examination of the kidney. The examiner strikes the patient's lumbar area with the ulnar side of his/her palm. Pain is a positive result and may be a sign of kidney inflammation (pyelonephritis).

The kidneys are located **in the retroperitoneal space** on both sides of the vertebral column. The left kidney extends from vertebra T12 to L2 and the right kidney extends from vertebra T12 to L3. There are **five renal segments**, defined according to their vascular supply. Ventrally, the kidneys are covered by parietal peritoneum. Their dorsal surfaces lie close to the muscles of the posterior abdominal wall. **The hilum of the left kidney lies against vertebra L1**. The hilum of the right kidney lies slightly more caudal than this, as it is pushed by the mass of the liver. At birth, the kidney is composed of 6 renal pyramids. This number increases to 7–18 in adulthood.

Coverings of the kidney

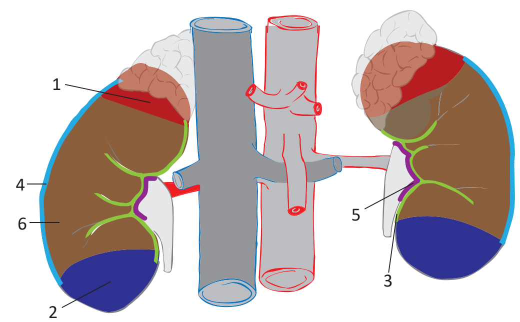
- 1 **Paranephric fat** (*corpus adiposum pararenale*) – a body of fat surrounding the kidney, located between the retrorenal lamina of the renal fascia and the transversalis fascia
  – extends caudally to the upper margins of the iliac fossa
- 2 **Renal fascia / Gerota's fascia** (*fascia renalis*) – is composed of an anterior and a posterior layer that fuse at the lateral margin and superior pole of the kidney
  – the layers remain separated caudally, creating a space that the kidney can move into (*ren migrans*)
  – contains the capsuloadipose vessels
  – represents a condensed extension of the transversal fascia, the internal lining of the abdominal cavity
  - 2.1 **Prerenal layer of Toldt** (*lamina prerenalis*) – the anterior layer
  - 2.2 **Retrorenal layer of Zuckerkandl** (*lamina retrorenalis*) – the posterior layer
- 3 **Perinephric fat** (*capsula adiposa*) – the fat capsule of the kidney, surrounded by the renal fascia
- 4 **Fibrous capsule** (*capsula fibrosa*) – loosely covers the kidney
  – is attached firmly only to the vessels and the renal pelvis at the hilum of the kidney



Transversal section of the right kidney at the level of L1 (vertebra)

External structures and segments

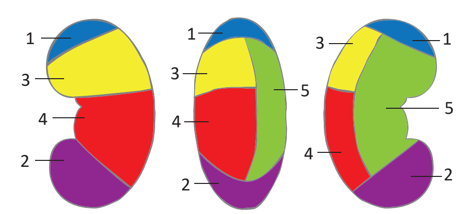
- 1 **Superior pole** (*extremitas superior*) – is covered by the suprarenal gland
- 2 **Inferior pole** (*extremitas inferior*)
- 3 **Medial border** (*margo medialis*) – the location of the hilum of the kidney
- 4 **Lateral border** (*margo lateralis*)
- 5 **Hilum of kidney** (*hilum renale*) – a vertical slit on the medial border where the renal vessels enter and leave the kidney and the location of the renal pelvis
- 6 **Anterior surface** (*facies anterior*)
- 7 **Posterior surface** (*facies posterior*)



Anterior view of the kidneys and great vessels

Segments

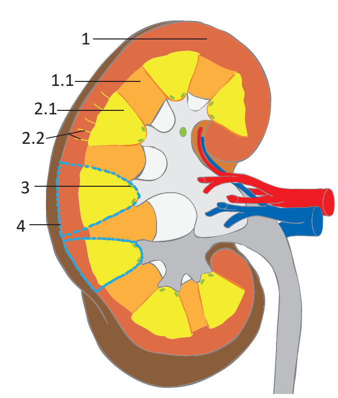
- 1 **Superior segment** (*segmentum superius*) – is supplied by the superior segmental artery *(a. segmenti superioris)*
- 2 **Inferior segment** (*segmentum inferius*) – is supplied by the inferior segmental artery (*a. segmenti inferioris*)
- 3 **Anterior superior segment** (*segmentum anterius superius*) – is supplied by the anterior superior segmental artery (*a. segmenti anterioris superioris*)
- 4 **Anterior inferior segment** (*segmentum anterius inferius*) – is supplied by the anterior inferior segmental artery (*a. segmenti anterioris inferioris*)
- 5 **Posterior segment** (*segmentum posterius*) – is supplied by the posterior segmental artery (*a. segmenti posterioris*)



Anterior, lateral, and posterior view of the left kidney

Internal structures

- 1 **Renal cortex** (*cortex renalis*) – is composed of the glomeruli and the proximal and distal tubules
  - 1.1 **Renal columns** (*columnae renales*) – extensions of the cortex into the renal medulla
- 2 **Renal medulla** (*medulla renalis*) – is composed of the intermediate tubules, tubules of the juxtamedullary nephrons and collecting ducts in renal pyramids
  - 2.1 **Renal pyramids** (*pyramides renales*) – their bases projecting outwards to the renal cortex
    – there are 7–18 renal pyraminds in the adult kidney
  - 2.2 **Renal medullary rays** (*radii medullares*)
    – slender strips of renal medulla extending into the cortex
- 3 **Renal papillae** (*papillae renales*) – round tips of the pyramids projecting into the hilum of the kidney
  - 3.1 **Cribriform area** (*area cribrosa*) – the perforated surface of the papilla
  - 3.2 **Openings of papillary ducts** (*foramina papillaria*)
    – through these openings, the collecting ducts empty in the minor calix
- 4 **Kidney lobes** (*lobi renales*) – are composed of the renal pyramids and contiguous cortex
  – macroscopically visible in the early periods of development (lobulated kidney)



Frontal section of the right kidney, anterior view

Histology

Nephron

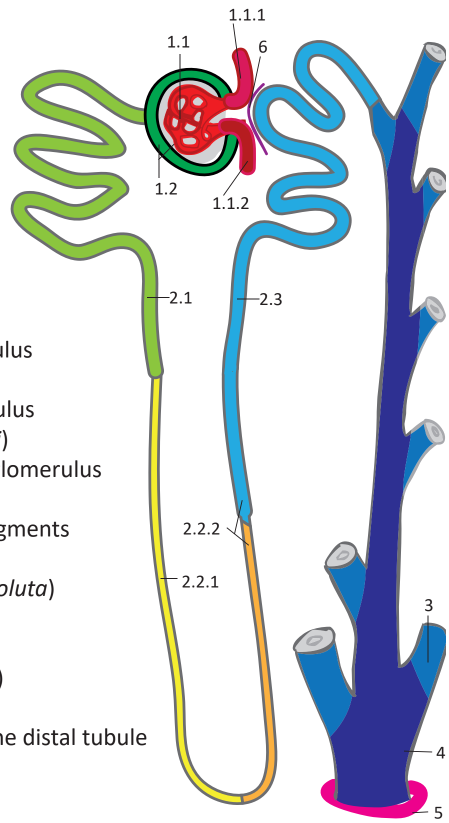
– the single functional unit of renal tissue
– **the cortical nephron** is located in the renal cortex
– **the juxtamedullary nephron** is located in the cortex in close proximity to the renal medulla

1 **Renal corpuscle** (*corpusculum renale Malpighi*)
– the initial part of the nephron
– consists of the glomerulus and Bowman's capsule

- 1.1 **Glomerulus** – a cluster of capillary loops
  - 1.1.1 **Afferent arteriole**
    – the larger vessel entering the glomerulus
  - 1.1.2 **Efferent arteriole**
    – the smaller vessel leaving the glomerulus
- 1.2 **Bowman's capsule** (*capsula glomeruli Bowmani*)
  – a double layer of epithelium surrounding the glomerulus
  – part of the renal filtration barrier

2 **Renal tubule** (*tubulus renalis*) – is divided into three segments

- 2.1 **Proximal tubule** (*tubulus proximalis*)
  - 2.1.1 **Proximal convoluted tubule** (*pars convoluta*)
  - 2.1.2 **Proximal straight tubule** (*pars recta*)
- 2.2 **Intermediate tubule** (*tubulus intermedius*)
  - 2.2.1 **Descending thin limb** (*crus descendens*)
  - 2.2.2 **Ascending thin limb** (*crus ascendens*)
    – the thick ascending limb is a part of the distal tubule
- 2.3 **Distal tubule** (*tubulus distalis*)
  – joins the collecting duct via the connecting tubule (*tubulus reuniens*)
  - 2.3.1 **Straight part** (*pars recta*)
  - 2.3.2 **Distal convoluted tubule** (*pars convoluta*)



Juxtamedullary nephron and intrarenal ducts

Intrarenal ducts

- 3 **Collecting ducts** (*ductus colligentes*) – originate close to the renal cortex
- 4 **Papillary ducts** (*ductus papillares*) – begin by a confluence of several collecting ducts
- 5 **Openings of papillary ducts** (*foramina papillaria*) – on the tips of the renal papillae through which urine flows into the minor calix

Special structure

- 6 **Juxtaglomerular complex** (*complexus juxtaglomerularis*) – an apparatus consisting of the juxtaglomerular cells (modified smooth muscle cells of the afferent and efferent arterioles), extraglomerular mesangium, macula densa of the distal tubule, and granular epithelial peripolar cells
  – maintains stable blood pressure and intravascular volume and regulates sodium and water homeostasis by releasing the hormone (enzyme) renin

Development

Ureteric bud (*diverticulum ureteris*) – originates from the mesonephric duct (*ductus mesonephricus Wolffi*)
– releases the signal for differentiation of surrounding mesodermal cells into nephrons
– without this stimulating structure, kidney tissue does not develop at all (renal aplasia)
– partial failure results in renal cystic diseases (cysts compress the healthy kidney tissue which leads to renal atrophy or even renal failure)

Function

1. **Plasma filtration** through the renal filtration barrier
2. **Secretion and resorption** of water, ions and other substances
3. **Antidiuretic hormone activity** in the collecting tubules
4. **Aldosterone activity** in the distal tubule
5. **Erythropoietin, renin and urodilatin production**
6. **Vascular portal system:** plasma filtration through the glomerular membrane (1st capillary system), absorption and secretion of water, ions and other substances in the peritubular capillary plexus (2nd capillary system)
7. **Hydroxylation of vitamin D**

The loop of Henle *(ansa nephroni)* is composed of the straight part of the proximal tubule, all of the intermediate tubule and the straight part of the distal tubule. The juxtamedullary nephrons have longer loops of Henle, whereas the cortical nephrons have shorter loops.

Clinical notes

Renal artery stenosis is a narrowing of the renal artery. Decreased blood flow in the renal vascular bed disrupts the regulation of blood pressure and can result in hypertension.

Brodel's bloodless line is a watershed area between areas supplied by the anterior and posterior branches of the renal artery. It carries a lower risk of bleeding when being cut during nephrotomy or lithotomy (surgical removal of kidney stones). Due to its low perfusion, it is an optimal place for insertion of a renal catheter.

The accessory renal artery (hilar artery) enters the hilum of the kidney along with the main renal artery. An aberrant accessory renal artery (polar artery) enters the kidney outside the hilum. It pierces the renal capsule at the inferior pole or superior pole. It more frequently enters the kidney at the inferior pole, where it may compress the transition zone between the renal pelvis and ureter and lead to hydronephrosis. Accessory renal arteries occur in 30 % of patients. During transplantation or nephrectomy, they may be a source of unexpected bleeding or ischemia due to the absence of arterio-arterial anastomosis within the kidney.

A floating kidney *(ren migrans / nephroptosis)* describes a situation in which the renal capsule is weakened due to an excessive loss of perinephric fat (extreme weight loss), and the kidney descends within the retroperitoneal space to the pelvic cavity. Although the renal vessels descend along the kidney, their origin remains at the level of vertebra L2.

An ectopic kidney is a kidney located in an abnormal position. It results from failure of the kidney to reach its normal position during development and is usually located more caudal than the normal position. Its arteries arise from the aorta at the corresponding level.

A transplanted kidney is usually positioned in the right iliac fossa. It is more common than an ectopic or floating kidney in the same location.

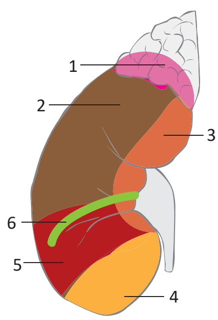
Regional anatomy

Anterior surface – right kidney
- 1 Suprarenal (adrenal) gland
- 2 Liver
- 3 Duodenum
- 4 Jejunum
- 5 Right colic flexure
- 6 Attachment of the transverse mesocolon

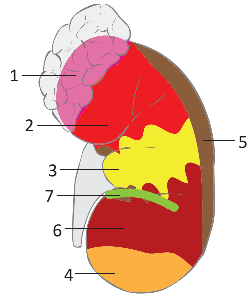
Anterior surface – left kidney
- 1 Suprarenal (adrenal) gland
- 2 Stomach
- 3 Pancreas
- 4 Jejunum
- 5 Spleen
- 6 Left colic flexure
- 7 Attachment of the transverse mesocolon

Posterior surface – same in both kidneys
- 1 Diaphragm
- 2 Psoas major
- 3 Quadratus lumborum
- 4 Transverse abdominal

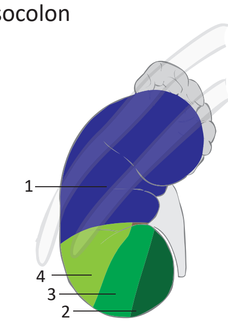
Structures related to the posterior surface of both kidneys:
- 11th and 12th rib
- subcostal nerve
- iliohypogastric nerve
- ilioinguinal nerve



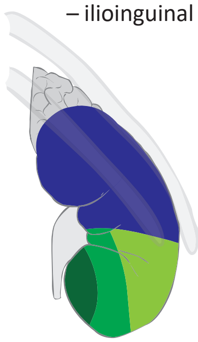
Right kidney



Left kidney

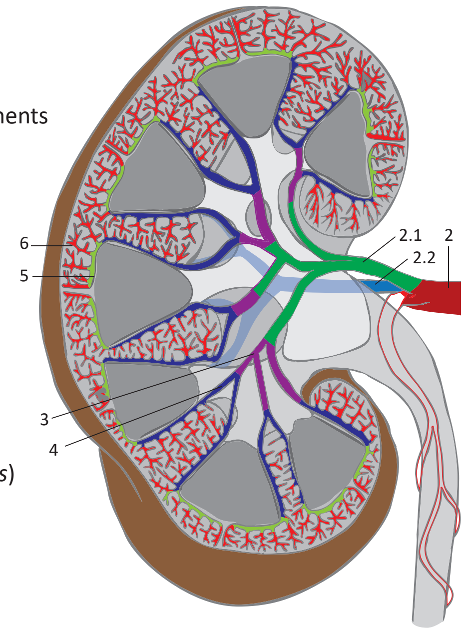


Left kidney


Right kidney

Vascular supply

Renal artery and its branches:
1 **Abdominal aorta** (*aorta abdominalis*)
2 **Right/left renal artery** (*arteria renalis dextra/sinistra*)
 2.1 **Anterior branch** (*ramus anterior*) – has 4 branches that supply the anterior segments
  – superior segmental artery (*a. segmenti superioris*)
  – inferior segmental artery (*a. segmenti inferioris*)
  – anterior superior segmental artery (*a. segmenti anterioris superioris*)
  – anterior inferior segmental artery (*a. segmenti anterioris inferioris*)
 2.2 **Posterior branch** (*ramus posterior*) – a single branch for the posterior segment
  – posterior segmental artery (*a. segmenti posterioris*)
3 **Lobar arteries** (*arteriae lobares*) – 15–20 arteries, one for each lobe
4 **Interlobar arteries** (*arteriae interlobares*)
 – pass through the renal cortex between the renal pyramids
5 **Arcuate arteries** (*arteriae arcuatae*) – always two
 – run between the border between the cortex and medulla
6 **Interlobular arteries, cortical radiate arteries** (*aa. corticales radiatae, aa. interlobulares*)
7 **Afferent glomerular arteriole** (*arteriola glomerularis afferens*)
8 **Glomerular capillary plexus** (*rete capillare glomerulare*)
9 **Efferent glomerular arteriole** (*arteriola glomerularis efferens*)
10 **Peritubular capillary plexus** – around proximal and distal tubules
11 **Vasa recta** (*arteriolae rectae*) – supply the medulla (juxtamedullary nephrons)

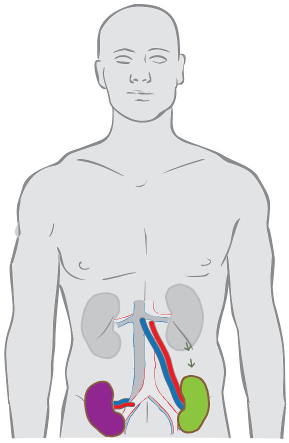


Coronal section of the right kidney, anterior view

Vascular portal system:
– **glomerular capillaries** continuing to the **peritubular capillary plexus or vasa recta**

Renal veins:
1. **Stellate veins** (*venulae stellatae*)
2. **Peritubular capillary plexus**
3. **Interlobular veins** (*venae interlobulares*)
4. **Arcuate veins** (*venae arcuatae*)
5. **Interlobar veins** (*venae interlobares*)
6. **Right/left renal vein** (*vena renalis dextra/sinistra*)
7. **Inferior vena cava** (*vena cava inferior*)

Lymphatic drainage: lateral aortic nodes, where three lymphatic plexuses terminate (one from the parenchyma, one from the fibrous capsule and one from the fat capsule)


- Ectopic kidney
- Floating kidney

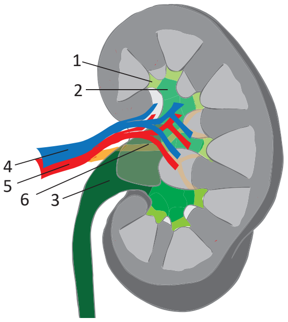
Innervation

Sympathetic, parasympathetic and visceral sensory innervation: renal plexus
Sympathetic supply: lesser splanchnic nerve (*n. splanchnicus minor*), coeliac ganglion (*ggl. coealiacum*)
Parasympathetic supply: vagus nerve (*n. vagus*)

The renal pelvis and renal calices are parts of the upper urinary tract. They **collect urine** from the kidney **and direct it into the ureters**. They have the same innervation, vascular supply and lymphatic drainage as the kidneys.

Structure

- 1 **Minor calices** (*calices minores*) – small calices encircling the renal papillae (7–18)
- 2 **Major calices** (*calices majores*) – formed by the fusion of 2–3 minor calices
- 3 **Renal pelvis** (*pelvis renalis*)

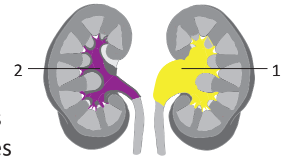


Anterior view of a coronal section of the left kidney

Regional anatomy of the hilum of the kidney

In ventrodorsal direction:

- 4 **Renal vein**
- 5 **Renal artery** – four anterior segmental branches
- **Renal pelvis**
- 6 **Renal artery** – posterior segmental branch



Types of the renal calices

- 1 **Ampullary type** – is distinguished by a wide pelvis and short calices
- 2 **Branching type** – is distinguished by a narrow pelvis and long calices

4 Ureter

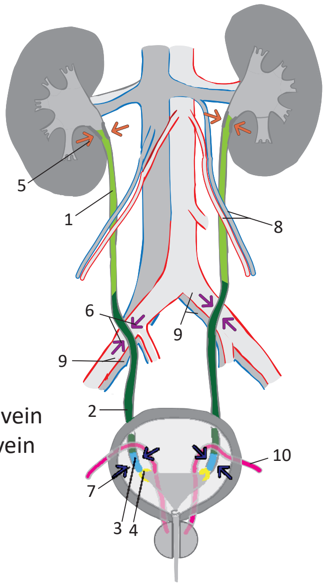
The ureter is a muscular tube that transports urine from the renal pelvis to the urinary bladder. Its **peristaltic movements** pass urine in waves to the urinary bladder. Therefore, urine flow is not continuous.

Parts and orifices

- 1 **Abdominal part** (*pars abdominalis*) – courses in the retroperitoneal space
- 2 **Pelvic part** (*pars pelvica*) – courses in the lesser pelvis
- 3 **Intramural part** (*pars intramuralis*) – passing through the wall of the urinary bladder
- 4 **Ureteric orifice** (*ostium ureteris*) – a slit-like orifice opened only during the flow of urine

Constrictions

- 5 **Junction with the renal pelvis**
- 6 **Crossing the iliac vessels**
  - – the right ureter runs ventrally to the external iliac artery and vein
  - – the left ureter runs ventrally to the common iliac artery and vein
- 7 **Intramural part** in the wall of the urinary bladder



Regional anatomy

The ureter courses:

- 8 Dorsally to the testicular/ovarian artery and veins
- 9 Ventrally to the common and external iliac artery and vein and the genitofemoral nerve
- 10 Dorsally to the ductus deferens / uterine artery

Vascular supply and innervation

Arterial supply
in both sexes: renal artery, internal iliac artery, superior vesicular artery and middle rectal artery
in males: plus the testicular artery and artery to the ductus deferens
in female: plus the ovarian and uterine arteries
Innervation: sympathetic, parasympathetic and visceral sensory nerve fibres from the ureteric plexus (*plexus uretericus*), which originate in the abdominal aortic and renal plexuses (*plexus aorticus abdominalis, plexus renalis*)

A mneumonic for the structures at the hilum of the kidney, in a ventrodorsal direction:
VAPA – vein, artery (renal), pelvis, artery (posterior segmental).

The ureter is 25–30 cm long and 4–7 mm wide.

The muscular layer (*tunica muscularis*) of the ureteric wall is composed of two layers: the internal longitudinal and external circular layer. The intramural part of the ureter has a ‘‘third collar-like’’ layer, called Waldeyer´s sheath.

The urachus is a remnant of the allantoenteric diverticulum. It is found within the median umbilical ligament (*ligamentum umbilicale medianum*) on the posterior side of the anterior abdominal wall.

The uretero-vesical junction region is described as the intramural part of the ureter, adjacent wall of the urinary bladder and ureteric orifice.

Clinical notes

In the endoscopic percutaneous approach to the renal calices, a needle is inserted in a direction parallel to the course of the calyx and directed towards the papilla. In this location the vasculature is most scarce, which eliminate the risk of bleeding.

Hydronephrosis describes a situation where the renal pelvis is dilated due to either obstruction in the flow of urine or upstream flow of urine from the urinary bladder to the ureters and renal pelves and calices.

Vesicoureteral reflux occurs when the uretero-vesical junction is insufficient, allowing urine to flow upstream, back into the ureters and renal pelvis.

Developmental anomalies of the ureter:
Ureter fissus – a ureter divided in two in its proximal part with a single opening into the urinary bladder.
Ureter duplex – a double ureter with two separate openings into the urinary bladder.

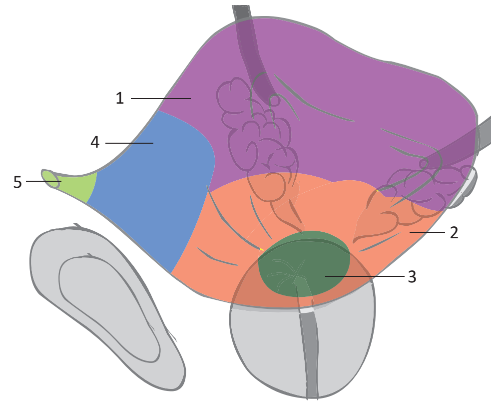
Ureterocele is a dilation of the distal part of the ureter curved inwards to the cavity of the urinary bladder.

The retropubic space of Retzius *(spatium retropubicum)* is used as an extraperitoneal surgical approach to the urinary bladder or prostate. It avoids entering the peritoneal cavity.

The urinary bladder is a subperitoneally located hollow organ. It **collects urine**, which it releases by contractions of the **detrusor muscle** *(m. detrusor vesicae)* into the urethra. As the amount of urine in the bladder increases, the tension in the walls of the bladder increases, which eventually leads to the activation of the sacral parasympathetic system via visceral sensory nerves. This triggers contractions of the detrusor muscle, leading to an urgent need to micturate.

External structures

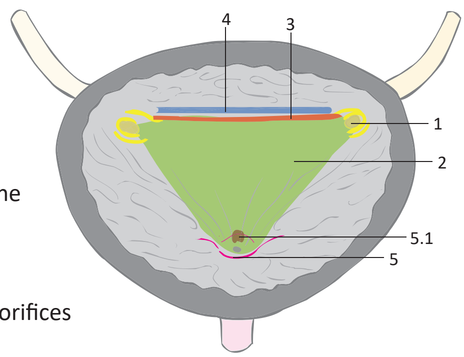
- 1 **Body of bladder** (*corpus vesicae*) – is covered by peritoneum
- 2 **Fundus of bladder** (*fundus vesicae*) – the bottom part of the bladder
  – in males, it is in close proximity to the prostate and rectum
  – in female, it is close to the vagina
- 3 **Neck of bladder** (*cervix vesicae*)
  – is located on the inferior surface of the bladder and is continuous with the urethra
  – in males, it prevents semen from flowing into the bladder during ejaculation
- 4 **Apex of bladder** (*apex vesicae*) – directed towards the pubic symphysis
  – in the empty bladder, it is located right behind the pubic symphysis
  – in the full bladder, it usually extends above the pubic symphysis
- 5 **Median umbilical ligament** (*ligamentum umbilicale medianum*)
  – a ligament extendingfrom the apex of the bladder to the umbilicus on the anterior abdominal wall between the transversal fascia and parietal peritoneum



Superior oblique lateral view of the urinary bladder

Internal structures

- 1 **Ureteric orifices** (*ostia ureterum*) – slit-like openings for the ureters in the bladder wall
  – there are no sphincter muscles around these orifices
- 2 **Trigone of bladder** (trigonum vesicae)
  – a triangle-like area in the infero-posterior part of the bladder
  – the ureteric orifices and the internal urethral orifice form the three corners of the trigone
  – its mucosa is not folded
  – in female, it projects on the anterior wall of the vagina as the trigonal area of Pawlik (*area trigonalis Pawliki*)
- 3 **Interureteric crest** (*plica interureterica*) – a fold in the bladder wall between the ureteric orifices
- 4 **Retrotrigonal fossa** (*fossa retrotrigonalis*) – a hollow space behind the interureteric crest
- 5 **Internal urethral orifice** (*ostium urethrae internum*) – an internal opening of the urethra
  - 5.1 **Uvula of bladder** (*uvula vesicae*)
    – a mucous fold in the posterior part of the internal urethral orifice
    – in males, it is located close to the middle lobe of the prostate

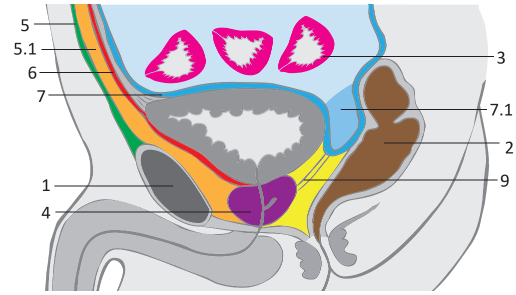


Oblique section of the urinary bladder, seen from the anterior superior view

Regional anatomy

Structures

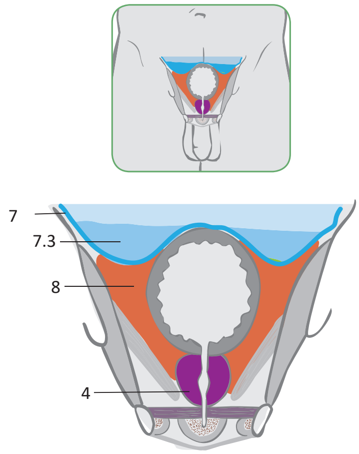
- 1 **Anterior:** pubic symphysis
- 2 **Posterior:** rectum in males, vagina in female
- 3 **Superior:** small intestine in the peritoneal cavity, uterus in female
- 4 **Inferior:** prostate in males, urogenital diaphragm in females



Sagittal section of the male pelvis

Spaces, fasciae, septa

- 5 **Transversal fascia** (*fascia transversalis*) – anterior to the apex of the urinary bladder
  - 5.1 **Retropubic space / space of Retzius** (*spatium retropubicum Retzii*)
    – a space between the urinary bladder and pubic symphysis
- 6 **Vesicoumbilical fascia of Delbet** (*fascia vesicoumbilicalis Delbeti*) – a fascia located between the medial umbilical ligaments and median umbilical ligament in the shape of a triangle
  – extends to the umbilicus and may contain the urinary bladder in its full state
- 7 **Parietal peritoneum** *(peritoneum parietale)* – covers the superior surface of the bladder
  - 7.1 **Recto-vesical pouch of Proust** (*excavatio rectovesicalis Prousti*)
    – the space between the rectum and urinary bladder (only in males)
  - 7.2 **Vesico-uterine pouch** (*excavatio vesicouterina*)
    – the space between the uterus and urinary bladder (only in females)
  - 7.3 **Paravesical fossae** (*fossae paravesicales*)
    – peritoneum-lined spaces on both sides of the urinary bladder
- 8 **Paravesical space** (*spatium paravesicale*) – a space surrounding the urinary bladder inside the pelvic cavity between the urinary bladder and pelvic walls
- 9 **Rectovesical septum of Denonvilliers** (*septum rectovesicale Denonvilliersi*)
  – a plate of connective tissue between the urinary bladder and the rectum (only in males)
  - 9.1 **Vesico-vaginal septum** (*septum vesicovaginale*) – an plate of connective tissue between the urinary bladder and the vagina with the adjacent cervix of the uterus



Frontal section of the male pelvis in the plane of the urinary bladder

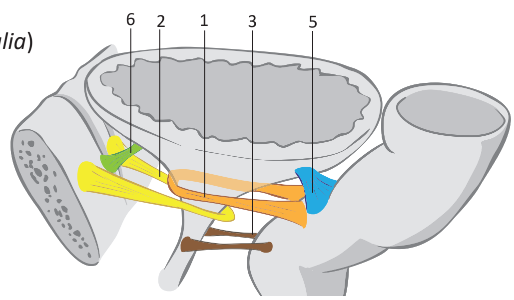
Fixation of the bladder

Smoooth muscles

- 1 **Rectovesicalis** (*musculus rectovesicalis*) – extends from the rectum to the anterior wall of the neck of the urinary bladder
- 2 **Pubovesicalis** (*musculus pubovesicalis*) – extends from the pubic rami to the posterior wall of the neck of the urinary bladder
- 3 **Rectourethralis** (*musculus rectourethralis*) – extends from the rectum to the urethra (only in males)
- 4 **Puboprostaticus** (*musculus puboprostaticus*) – extends from the pubic rami to the prostate

Ligaments

- 5 **Rectovesical ligaments** (*ligamenta rectovesicalia*) – extends from the rectum to the posterior wall of the urinary bladder
- 6 **Pubovesical ligaments** (*ligamenta pubovesicalia*) – extends from the pubic symphysis to the anterior wall of the urinary bladder
- 7 **Vesicouterine ligaments** (*ligamenta vesicouterina*) – extends from the urinary bladder to the uterus (only in females)
- 8 **Puboprostatic ligaments** (*ligamenta puboprostatica*) – extends from the pubic rami to the prostate (only in males)



Pubic symphysis, urinary bladder and rectum seen from the superior oblique left view

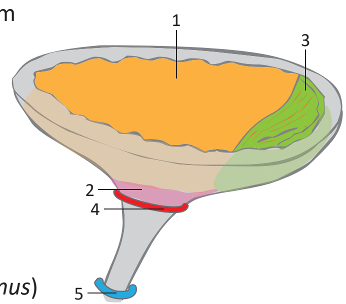
Muscles involved in micturition and continence

Urinary bladder muscles

- 1 **Detrusor (muscle)** (*musculus detrusor vesicae*) – three layers of smooth muscle (inner plexiform, middle circular and external longitudinal layer) innervated by the parasympathetic system
- 2 **Sphincter vesicae (muscle)** (*musculus sphincter vesicae*)
  – a single layer of circular smooth muscle extending around the neck of the bladder innervated by the sympathetic system
  – in male, prevents semen from flowing into the bladder during ejaculation
- 3 **Trigonal muscles** (*musculi trigoni vesicae*)
  – bundles of smooth muscle located in the trigone of bladder, organised in a loop-like fashion
  – this enables them to control the opening and closure of the internal urethral orifice and ureteral orifices

Muscles surrounding the urethra

- 4 **Internal urethral sphincter** (*musculus sphincter urethrae internus*) – the internal sphincter is composed of smooth muscle tissue
- 5 **External urethral sphincter** (*musculus sphincter urethrae externus*) – the external sphincter is composed of striated muscle tissue



Superior left oblique view of the urinary bladder and urethra

Vascular supply

Arteries:
- superior vesical arteries (*aa. vesicales superiores*) from the umbilical artery
- inferior vesical artery (*a. vesicalis inferior*) from the internal iliac artery
- vesical branches (*rr. vesicales*) from the surrounding arteries (obturator, inferior gluteal, uterine and vaginal arteries)

Veins:
- vesical venous plexus (*plexus venosus vesicalis*) via the vesical veins into the internal iliac vein
- connections with the prostatic/vaginal venous plexus and pudendal plexus of Santorini

Lymphatic drainage:
- paravesical nodes (*nodi lymphoidei paravesicales*), which drain to the common iliac, external and internal iliac nodes

Innervation

Sympathetic system: vesical plexus (from inferior hypogastric/pelvic plexus)
Parasympathetic system, visceral sensory and visceral motor nerves: sacral nerves (L2–S2)

Urokystis is the Greek term for urinary bladder.

The vertex is the highest point of a full urinary bladder.

Bell's muscular fascicles are a pair of muscular fascicles between the ureteric orifice and the internal urethral orifice.

The spongy layer of the female urethra becomes thinner with age and may contribute to incontinence. **Micturition** *(micturitio)* is the Latin term for urination.

Pawlik´s triangle *(area trigonalis vaginae)* is an area on the anterior wall of the vagina that is directly opposed to the trigone of the bladder. Vaginal rugae are absent here.

The external urethral sphincter is predominantly composed of slow--twitch muscle fibres that are capable of long-lasting contraction, maintaining closure of the urethra.

The mucosa and submucosa of the urethra are estrogen-dependent. They atrophy after menopause, which contributes to stress incontinence.

When the amount of urine in the urinary bladder reaches approximately 400 ml, the detrusor begins to contract, which creates the feeling of urgency. Volumes of urine over 500 ml can cause pain that projects to the hypogastrium, perineum and penis.

Clinical notes

The female urethra is short and in close proximity to the rectum. Females thus have an increased incidence of urinary tract infections caused by enteric pathogens.

The female urethra is short, wide and straight. Therefore, the insertion of a urinary catheter is not complicated.

The external iliac nodes need to be removed during radical cystectomy as they drain the urinary bladder.

The external urethral sphincter in males is the main mechanism for providing continence. During prostate and urinary bladder operations, it may be damaged and can cause urinary incontinence.

Epicystostomy *(sectio alta)* is a method to directly drain urine from the bladder by a inserting urine catheter through the anterior abdominal wall (extraperitoneal approach).

The female urethra is a 4 cm long, 6–8 cm wide muscular tube connecting the bladder to the external environment, thus forming the **terminal portion of the urinary tract in the female**. Voluntary contraction of the external urethral sphincter prevents urine from leaking out during reflex contraction of the detrusor muscle.

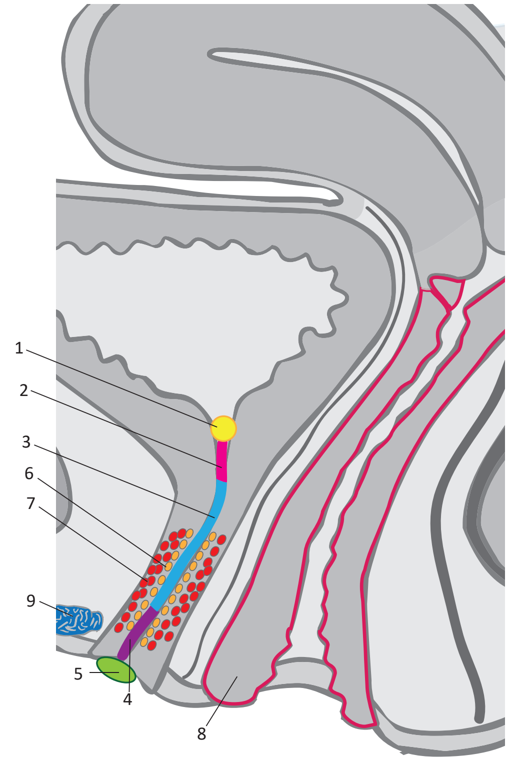
Parts

- 1 **Internal urethral orifice** (*ostium urethrae internum*) – is located inside the urinary bladder
- 2 **Intramural part** (*pars intramuralis*) – runs through the urinary bladder wall
- 3 **Pelvic part** (*pars pelvica*) – the longest section, located between the urinary bladder wall and the pelvic floor
- 4 **Perineal part** (*pars perinealis*) – passes through the perineal muscles and the urogenital hiatus (*hiatus urogenitalis*) of the levator ani
- 5 **External urethral orifice** (*ostium urethrae externum*)
  – protrudes as the urethral papilla (*papilla urethralis*) in the vestibule (*vestibulum vaginae*)

Internal structure

Mucosa (*tunica mucosa*) – lined by stratified squamous nonkeratinized epithelium
Spongy layer (*tunica spongiosa*) – part of the mucosa
– contains a venous plexus, which is essential for maintenance of urinary continence
Muscular layer (*tunica muscularis*)
– contains two layers of muscle: an inner longitudinal layer and an outer circular layer
– contains two spincters:

- 6 **Internal urethral sphincter** (*musculus sphincter urethrae internus*)
  – a weak layer of smooth muscle
- 7 **External urethral sphincter** (*musculus sphincter urethrae externus*)
  – a layer of striated muscle



Left sagittal view of the female pelvis with the female urethra enlarged

Regional anatomy

- 8 **Vagina** – the urethra lies on the anterior wall of vagina, on which it forms a crest called the urethral carina of the vagina (*carina urethralis vaginae*), which is an extention of the trigonal area of Pawlik
- 9 **Bulb of vestibule** (*bulbus vestibuli*) – in front of and on both sides of the urethra

Vascular supply and innervation

Arteries:
– **intramural part:** inferior vesical artery (from the internal iliac artery)
– **pelvic part:** uterine and vaginal arteries (from the internal iliac artery)
– **perineal part:** internal pudendal artery (from the internal iliac artery)
Veins:
– **intramural part:** vesical venous plexus (via the vesical veins to the internal iliac vein)
– **pelvic part:** vaginal venous plexus (via the vaginal veins to the internal iliac vein)
– **perineal part:** internal pudendal veins (to the internal iliac vein)
Lymphatic drainage: paravesical nodes (to the common, internal and external iliac nodes)
Sympathetic and parasympathetic, visceral sensory and (somatic) motor nerve fibers: vesical plexus

7 Dynamics of micturition

Urinary continence is the ability to keep urine in the bladder and excrete it in a controlled fashion. **Urinary incontinence** is the lack of this ability. The two main aspects of urinary continence are the position of the urethra and neck of the bladder and the proper function of the urethral sphincters. In females, the sphincter compresses the urethra against the vagina and thus closes it. **Stress incontinence** is defined as involuntary leakage of urine when the pressure inside the peritoneal cavity rises (e.g. during coughing or laughing). Increased intra-abdominal pressure is able to push the urethra and the adjacent neck of the bladder downwards (a condition termed urethral hypermobility). Pressure is thus distributed unevenly; it is higher inside the urinary bladder than in the urethra, which results in urine leakage. **A different kind of stress incontinence** occurs when the urethra and neck of the bladder are permanently descended into the lesser pelvis, even at rest. Consequently, the internal urethral sphincter becomes unable to keep urine from escaping the urinary bladder.

Urinary continence in male is provided by:
– urethral mucosa and submucosa
– smooth muscle of the urethra
– external urethral sphincter
– levator ani (pubourethralis)

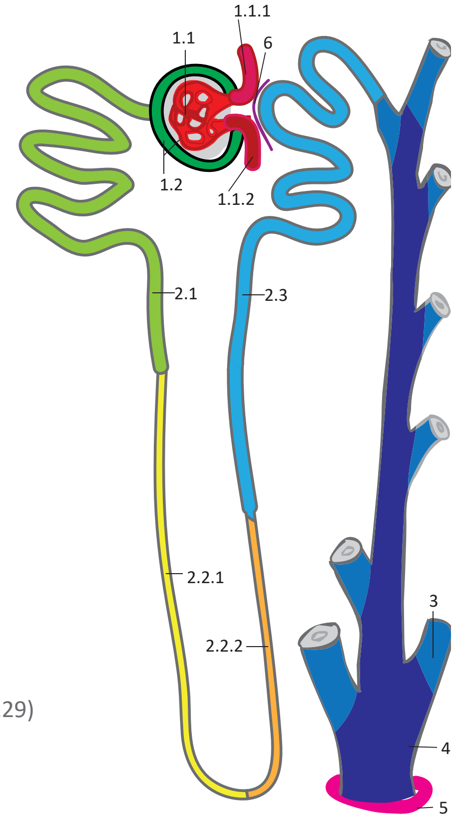
Urinary continence in female is provided by:
– urethral mucosa and submucosa
– smooth and striated muscle of the urethra
– external urethral sphincter, compressor urethrae
– sphincter urethrovaginalis, levator ani (pubovaginalis)

I. General overview and development of the urinary system

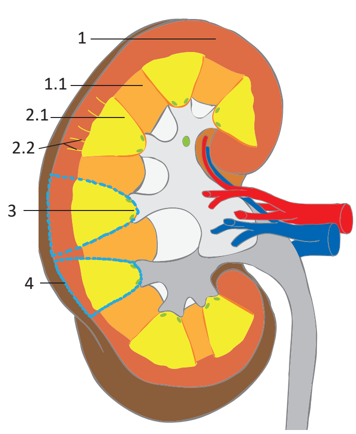
1. Name three structures that develop from the ureteric diverticulum. (p. 226)
2. Name the longest segment of the urinary tract. (p. 226)
3. Describe the structure of the three layers of the wall of tubular organs of the urinary system. (p. 226)
4. Specify which layer of the urinary bladder forms the fold between the urinary bladder and uterus. (p. 226)
5. Identify the difference between the mesonephros and metanephros. (p. 226)
6. Specify the definitive structure the mesonephric duct develops into (in the male). (p. 226)
7. Name the common structure from which the urinary bladder, prostatic and membranous parts of the urethra develop. (p. 226)

II. Kidneys

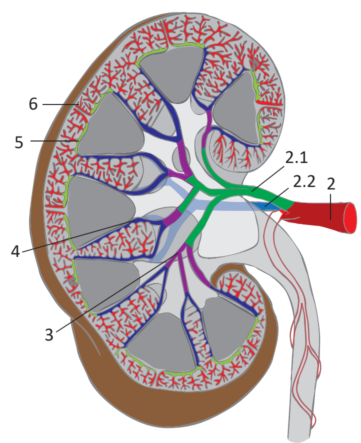
8. Name the 4 layers of the renal capsule. (p. 227)
9. Identify the 2 layers of the renal fascia of Gerota. (p. 227)
10. Explain the definition of a renal segment. (p. 227)
11. Draw the anterior and posterior views of renal segments. (p. 227)
12. Define a renal lobe. (p. 227)
13. Explain under which circumstances a lobulated kidney is physiologically present. (p. 227)
14. Describe the strucutre of the renal columns. (p. 227)
15. Explain the function of the juxtaglomerular complex. (p. 228)
16. Name the 3 segments of the intrarenal excretory system. (p. 228)
17. Name the parts of the nephron. (p. 228)
18. Explain the difference between the cortical and juxtaglomerular nephron. (p. 228)
19. State the 6 main functions of the kidney. (p. 228)
20. Draw and explain the regional anatomy of the kidney's anterior and posterior surfaces. (p. 229)
21. Describe a scheme of the vasculature of the kidney. (p. 229)
22. Explain the lymphatic drainage of the kidney. (p. 229)
23. State the structure that is continuous with the renal collecting ducts on one side and renal pelvis on the other. (p. 229)



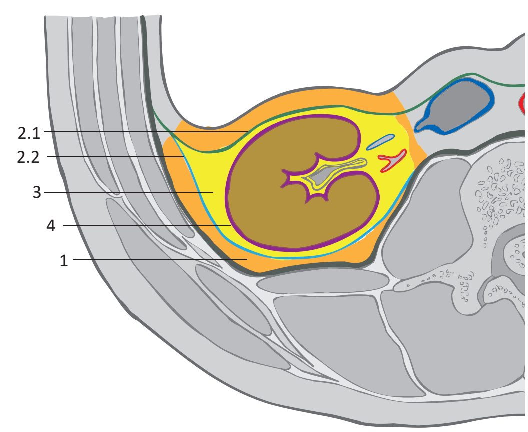
Describe the structure of the nephron and intrarenal ducts



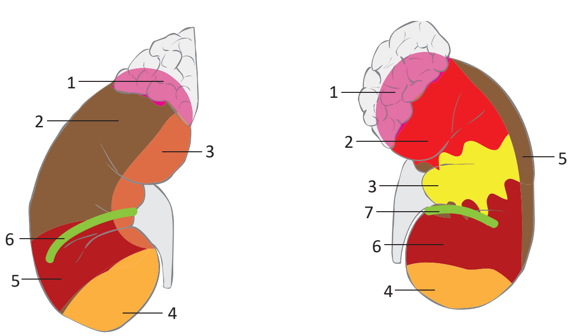
Describe the internal structure of the kidney



Describe the vasculature of the kidney



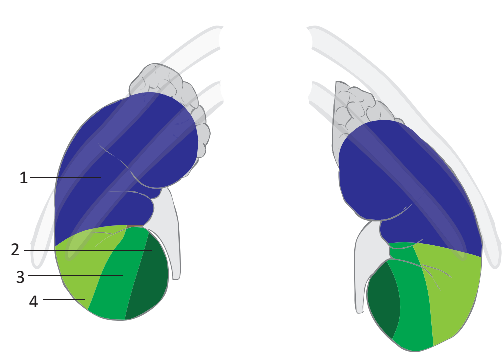
Describe the renal fasciae



Right kidney

Left kidney

Describe the regional anatomy of the kidneys from its anterior aspect



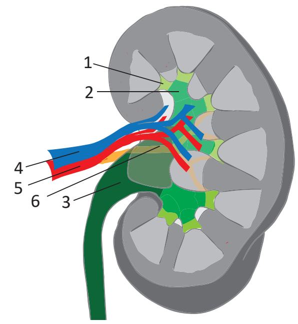
Left kidney

Right kidney

Describe the regional anatomy of the kidneys from its posterior aspect

III. Renal calices and the renal pelvis

24. Identify the 2 main types of renal calices and explain the difference between them. (p. 230)
25. Name the 4 structures of the hilum of the kidney. (p. 230)
26. Identify the structure that is posterior to the renal pelvis in the hilum of the kidney. (p. 230)
27. Identify the most anterior structure in the hilum of the kidney. (p. 230)



Describe the renal calyces, renal pelvis and structures in the hilum of the kidney

IV. Ureter

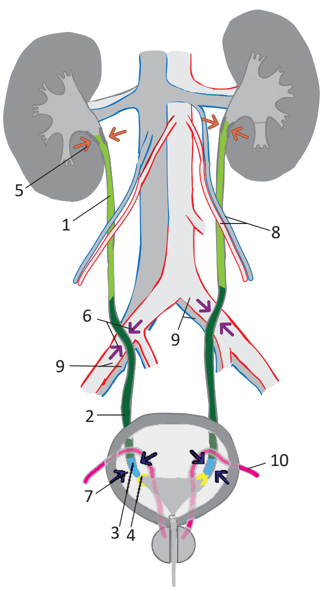
28. Define the segments of the ureter and orifice of the ureter. (p. 230)
29. Define the segments or the ureter that are the most vulnerable to obstruction (i.e. by kidney stones) due to their anatomical constriction. (p. 230)
30. Identify the 2 structures coursing anterior to the ureter. (p. 230)
31. Identify the structure coursing dorsally from the ureter. (p. 230)
32. Name the 3 vessels that supply the ureter via their ureteric branches. (p. 230)

V. Urinary bladder

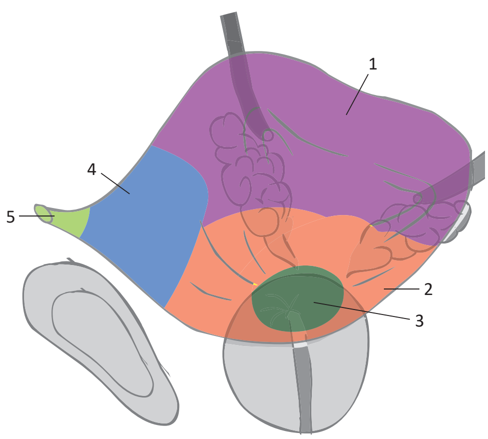
33. Name the 5 parts of the urinary bladder. (p. 231)
34. Describe the borders of the trigone of the bladder. (p. 231)
35. Define the location where urine may stagnate due to an enlarged prostate. (p. 231)
36. Draw the sagittal view of the male pelvis and describe the regional anatomy of the urinary bladder. (p. 231)
37. Identify the ligaments that contribute to the fixation of the urinary bladder. (p. 232)
38. Distinguish the structures that contribute to fixation of the urinary bladder specific for male/female. (p. 232)
39. Name the muscles of the urinary bladder and the muscles in close proximity to the urethra that contribute to urinary continence in both males and females. (p. 232)

VI. Female urethra

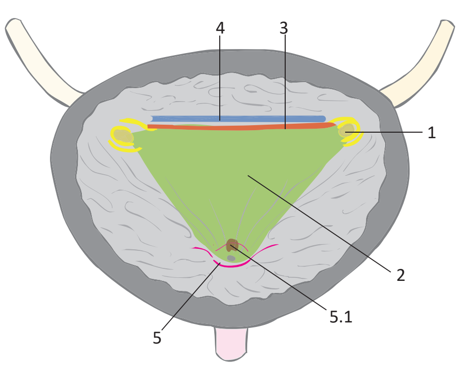
40. Define the 5 segments of the female urethra. (p. 233)
41. Explain the reason why females are more vulnerable to urinary infections. (p. 233)



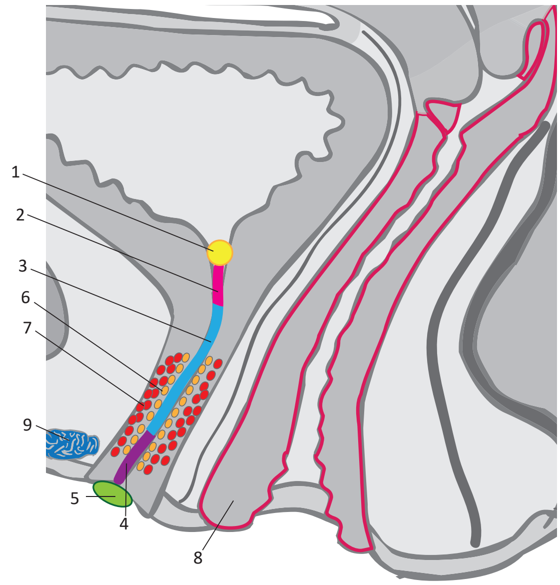
Describe the course of the ureter



Describe the external structure of the urinary bladder



Describe the internal structure of the urinary bladder



Describe the structure of the female urethra

We would like to **thank the following anatomists, clinicians and medical students** for their invaluable help, devotion and feedback in the preparation of this chapter.

Anatomists

Assoc. prof. Václav Báča, MD, PhD – Department of Anatomy, Third Faculty of Medicine, Prague, Czech Republic
Assoc. prof. Adriana Boleková, MD, PhD – Department of Anatomy, Pavol Jozef Šafárik University, Košice, Slovakia
Assoc. prof. Veronica Macchi, MD PhD – Institute of Human Anatomy, University of Padova, Italy

Clinicians

Dale Kalina, MD – Department of Internal Medicine, University of Saskatchewan, Canada

Medical students

Klára Macháčková
Danil Yershov
Eva Fürstová
Petr Urban

9 References

1. AGUR, A. M. R. and A. F. DALLEY. Grant's Atlas of Anatomy. 13th Ed. Philadelphia: Lippincot Williams & Wilkins, 2012, 888 p. ISBN 978-1-60831-756-1.
2. HALL, J. E. Guyton and Hall Textbook of Medical Physiology. 12th Ed. Philadelphia: Saunders/Elsevier, 2011, 1120 p. ISBN 978-1-4160-4574-8.
3. CHUNG, K. W.; CHUNG, H. M. and N. L. HALLIDAY. BRS Gross Anatomy. 8th Ed. Philadelphia: Lippincott Williams & Wilkins, 2015. 544 p. ISBN 978-1-4511-9307-7.
4. LANZ, T. and W. WACHSMUTH. Praktische Anatomie. 7 bands. Berlin: Springer, 2003, 3658 p. ISBN 978-3-540-40571-9.
5. MILLS, S. E. Histology for pathologists. 4th Ed. Philadelphia: Lippincot Williams & Wilkins, 2012. 1328 p. ISBN-13: 978-1-4511-1303-7.
6. NETTER, F. H. Atlas of Human Anatomy. Professional Edition. 6th Ed. Oxford: Elsevier. 2014. 640 p. ISBN 978-1-455-75888-3.
7. OVALLE, W. K. and P. C. NAHIRNEY. Netter's Essential Histology: with Student Consult Access. 2nd Ed. Philadelphia: Saunders/Elsevier, 2013. 493 p. ISBN-13: 978-1-4557-0631-0.
8. SNELL, R. S. Clinical anatomy by regions. 9th Ed. Philadelphia: Lippincott Williams & Wilkins, 2012. 768 p. ISBN 978-1-60913-446-4.
9. WILLIAMS, P. L., ed. Gray´s Anatomy: the anatomical basis of medicine and surgery. 38th Ed. New York: Churchill Livingstone, 1995, 2092 p. ISBN 0-443-05717-6.